

Compose a Breakfast Tray SOP

One episode is one full composition of a breakfast tray on the tabletop station. It runs from an empty tray on its build spot, a folded napkin and two pieces of cutlery on a linen mat, and a plate, two condiment pots, and a cup filled with juice on a staging shelf, to a loaded tray standing square and fully inside the delivery outline with no liquid anywhere outside the cup.

The tray carries a printed liner, and the liner is the only thing that says where anything goes. It has four marks: the **napkin outline** across the left end, the **plate outline** in the center, the **cup outline** at the back right, and two **condiment circles** at the front right, one labelled jam and one labelled butter. Nothing on the liner holds an item in place. Every item is set down flat on its own mark, clear of its neighbours, and left there.

Arm roles are fixed and never swap. The **left gripper is the linen gripper**: the folded napkin, the fork, and the spoon. It works the linen mat and the left end of the tray. The **right gripper is the service gripper**: the plate, the two condiment pots, and the filled cup. It works the staging shelf and the center and right of the tray. Neither gripper reaches into the other gripper's side, nothing is ever handed from one gripper to the other, and only one arm is over the tray at a time. Their one shared action is Step 5, where both grippers slide the loaded tray together. That is the only step with no idle arm.

The order never changes: **napkin, cutlery, plate, condiments, then the cup, then the slide**. The napkin goes down before the cutlery that rests on it. Within a pair, the far mark is filled first, so the fork goes down before the spoon and the jam pot before the butter pot. **The cup is always last, and nothing ever passes over the cup once it is down.**

Never lift, turn, or push the tray while it is being loaded, and never lift it off the table during the slide. Never carry more than one item per trip, and never drag or slide an item across the liner: lift it clear and carry it. Never turn an item in the air to change the way it faces: stand it back on its staging spot and take it again. Never grip the cup by its rim or its handle, never tip the cup, and never top it up or pour out of it. Never peel, lift, or press a condiment lid. Never unfold, refold, or shake out the napkin. Never rest cutlery on the plate or on the bare liner. Never nudge an item into place after the gripper has opened.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the tray and its frame includes the build spot, the linen mat, the staging shelf, and the delivery zone.
3. All four liner marks, both condiment labels, the fill line on the cup, and the delivery outline are readable on camera.
4. Both arms are at home with grippers open.
5. The table holds nothing but the tray on its build spot, the linen mat with its three pieces, the staging shelf with its four items, and the bare delivery zone.
6. The left arm reaches the whole linen mat, the napkin outline, and the left tray rail at both the build spot and the delivery zone, without extending to a joint limit, and reaches nothing on the right side of the table.
7. The right arm reaches the whole staging shelf, the plate outline, the cup outline, both condiment circles, and the right tray rail at both the build spot and the delivery zone, without extending to a joint limit, and reaches nothing on the left side of the table.
8. Each arm comes onto the tray straight down from above, without reaching past the back table edge, and both grippers stand on their own rail at the same time without the arms fouling each other.
9. The path from the build spot to the delivery zone is clear of everything, and neither arm collides with the tray on the way into the center. If a zone cannot be reached, move it toward the arm that cannot reach it until lines 6 to 8 hold.

Materials checklist

1. The tray stands empty, level, and flat on the build spot, squared to the back table edge, with its long side running left to right.
2. The tray has a raised lip all round and one grip rail at each end, and each rail is deep enough for a gripper to close on it with the tray flat on the table.
3. The tray slides across the bare table under a straight push, without catching or tipping, and stays put when an item is set down on it straight from above. It slides over the delivery outline without catching a taped edge, and the tape stays stuck down and flat.
4. The printed liner lies flat inside the tray, fully under the lip, and does not slide or lift when an item is set on it.
5. The liner carries exactly four marks and nothing else: the napkin outline across the left end, the plate outline in the center, the cup outline at the back right, and two labelled condiment circles at the front right, jam on the left and butter on the right.
6. Every mark is large enough that its item sits fully inside it, and the marks are far enough apart that no two items touch when each sits on its own mark.

7. The staging shelf sits to the right of the tray and holds the plate lying flat, the jam pot and the butter pot standing upright, and the cup standing upright with its handle pointing to the right.
8. The plate is dry, is not chipped, sits flat on a bare surface without rocking, and fits fully inside the plate outline.
9. Both condiment pots are the same size, are sealed with an unbroken lid, stand upright unaided on a bare flat surface, and can each be lifted by one gripper on the body clear of the lid.
10. The cup holds juice level with its fill line, has no lid, has a handle clear of the body so a gripper closes on the body without touching the handle, and stands upright unaided without rocking when it is full.
11. The fill line and the juice surface are both visible from the front.
12. The linen mat sits to the left of the tray and holds the folded napkin lying flat, with the fork and the spoon lying beside it, each on its own spot and not touching.
13. The napkin is already folded into a flat rectangle, is dry, holds its fold when it is lifted by one edge, and fits fully inside the napkin outline.
14. The fork and the spoon are clean and dry, lie flat on a bare surface without rocking, and both fit fully on the folded napkin side by side without touching.
15. The delivery outline is marked directly on the bare tabletop at the center front, in flat tape laid flush with the table, and the tray fits fully inside it. Nothing sits on it or inside it, and there is no mat, cloth, or liner of any kind between the build spot and the delivery outline.

Workspace layout

There is no wall behind the station. Nothing is marked or taped out apart from the liner marks and the delivery outline. Every area below is a fixed place on the bare tabletop, judged by eye against the table edges and the tray itself.

- **Back table edge:** the far edge of the table. The tray is squared to it at the build spot and nothing overhangs it.
- **Front table edge:** the near edge of the table. The tray is squared to it once it reaches the delivery zone.
- **Build spot:** the area at the center back of the table where the tray sits while it is loaded. The tray straddles the table midline here, so the left gripper reaches its left end and the right gripper reaches its center and right end.
- **Tray:** the open flat tray with a raised lip and two end rails. It stays flat on the table for the whole episode. It is never lifted, and it moves only once, in Step 5.
- **Left rail** and **right rail:** the grip rails at the two ends of the tray. The left gripper takes the left rail, the right gripper takes the right rail, and only in Step 5.

- **Liner:** the printed sheet lying flat inside the tray, carrying the four marks. It is never lifted or moved.
- **Napkin outline:** the mark across the left end of the liner. It holds the folded napkin, and the fork and spoon rest on the napkin.
- **Plate outline:** the mark in the center of the liner. It holds the plate.
- **Cup outline:** the mark at the back right of the liner. It holds the filled cup, and it is the last mark filled.
- **Condiment circles:** the two labelled circles at the front right of the liner, jam on the left and butter on the right.
- **Linen mat:** the mat to the left of the tray holding the folded napkin, the fork, and the spoon at episode start, empty at episode end. Left gripper only.
- **Staging shelf:** the shelf to the right of the tray holding the plate, both condiment pots, and the filled cup at episode start, empty at episode end. Right gripper only.
- **Delivery zone:** the area at the center front of the table carrying the delivery outline, taped flat on the bare tabletop. It is bare at episode start and holds the loaded tray at episode end. Both grippers reach it without extending.
- **Arm home:** the resting pose for each arm, gripper open and clear of the table.

Vocabulary

- **Arm assignment:** fixed by object, not by where the object lies. The **left gripper** handles the folded napkin, the fork, and the spoon, and works the linen mat and the left end of the tray. The **right gripper** handles the plate, both condiment pots, and the cup, and works the staging shelf and the center and right of the tray. In Step 5 the left gripper takes the left rail and the right gripper takes the right rail. Neither gripper touches the other gripper's items at any point, and nothing is handed between grippers.
- **Idle arm:** the arm not named in a step. It waits at its home, clear of the table, for the whole of that step. It never steadies an item, holds the napkin down, holds the tray, or assists the working arm. **Step 5 is the only step with no idle arm.**
- **Sliding together:** both grippers hold their own rail and move as one. They close at the same time, start and stop together, travel at the same speed, and keep the tray the same distance from the front table edge at both ends, so the tray stays square all the way. Neither gripper leads, drags, or moves its end on its own.
- **Item:** any object a gripper handles: the napkin, the fork, the spoon, the plate, a condiment pot, or the cup.
- **Cutlery:** the fork and the spoon. They rest on the folded napkin and nowhere else.
- **On its mark:** the item lies or stands fully inside its own liner mark, flat and settled, with no part over the edge of the mark and no part touching a neighbouring item. It holds itself there:

it does not lean on a neighbour, does not rest on the lip of the tray, and does not overhang the tray.

- **Set for service:** the folded napkin lies flat inside the napkin outline with its long side running front to back, and the fork and the spoon lie flat on top of it, parallel to each other, handles toward the front table edge, fully on the napkin and not touching each other. The **fork** lies on the half of the napkin nearer the plate, the **spoon** on the half nearer the left rail.
- **Fill line:** the line marked on the cup. The juice stands level with it at episode start and is still level with it at episode end. The cup is never topped up and never poured out.
- **Level carry:** the cup is lifted straight up, carried at one height with its rim flat, and lowered straight down. The gripper does not tip it, swing it, or turn it, and the juice surface stays flat and still.
- **Spill:** any juice outside the cup, on the liner, the tray, an item, the staging shelf, or the table, and any condiment outside a pot. A wet ring on the liner is a spill.
- **Square on the delivery outline:** the tray sits fully inside the delivery outline, flat on the table, with its long side parallel to the front table edge and no part over the outline or over the table edge.
- **Release point:** the left gripper opens only over the linen mat or over the napkin outline. The right gripper opens only over the staging shelf, the plate outline, a condiment circle, or the cup outline. Neither ever opens over the bare liner, over another item, or over a mark the held item does not belong on.
- **Stable placement:** the item stays still after release, fully inside its own mark, and does not rock, roll, slide, tip, or lean on a neighbour.

Steps

Step 1: Set the napkin and the cutlery

Goal: the left end of the tray is set for service and the linen mat is empty.

The right arm is the idle arm for the whole of Step 1. Work **napkin, fork, then spoon**, one piece per trip, with the **left gripper** only. The fork goes down before the spoon because the fork lies on the far half of the napkin, so nothing is carried over a piece already down.

1.1 Lay the folded napkin

- The **left gripper** approaches the folded napkin on the linen mat from above and closes on one edge of it, square to the edge, so the napkin hangs flat and keeps its fold.
- Lift it straight up clear of the mat and carry it flat to a point directly above the napkin outline, with its long side running front to back.
- Lower it straight down until it lies flat on the liner, open the gripper, and lift clear.

Carry the napkin flat and keep the fold. Do not let it hang by a corner, do not ball it up, do not unfold or refold it, and do not shake it out.

Check: the napkin lies flat and folded, fully inside the napkin outline, with no corner turned under and no part over the lip of the tray. If it lands folded over on itself, off the outline, or crooked, lift it clear, lay it back on the linen mat, and lay it again once.

1.2 Place the fork

- Approach the fork on the linen mat from above and close on its handle, square to it so it hangs level.
- Lift it straight up clear of the mat and carry it level to a point directly above the half of the napkin nearer the plate outline, with its handle already pointing toward the front table edge.
- Lower it straight down until it lies flat on the napkin, open the gripper, and lift clear.

If it comes up pointing the wrong way, lay it back on the linen mat, let go, and take it again with the handle toward the front table edge. Never turn it in the air.

Check: the fork lies flat and still, fully on the napkin, handle toward the front table edge, and does not rest on the bare liner or on the lip of the tray. If it hangs off the napkin or lands crossed over the napkin, lift it clear and lay it down again once.

1.3 Place the spoon

- Take the spoon the same way, and lay it on the half of the napkin nearer the left rail, parallel to the fork, handle toward the front table edge.
- Lower it straight down, open the gripper, lift clear, and return the left arm to its home.

Check: the left end of the tray is set for service. The fork and the spoon lie parallel, do not touch each other, and both sit fully on the napkin. If they touch or cross, lift the spoon clear and lay it down again once. The linen mat is empty and the left gripper is at home before the right arm moves.

Step 2: Place the plate

Goal: the plate lies flat on its outline in the center of the tray.

The left arm is the idle arm for the whole of Step 2. The plate is the widest item on the tray, so it goes down before the condiments and the cup.

- The **right gripper** approaches the plate on the staging shelf from above and closes on its rim, square to it so the plate hangs level.
- Lift it straight up clear of the shelf and carry it level to a point directly above the plate outline, keeping it clear of the napkin and the cutlery already set at the left end.
- Lower it straight down until it lies flat on the liner, hold it a moment until it is clearly resting on its own, then open the gripper and lift clear.

Carry the plate level and set it down flat. Do not tip it on edge, do not slide it across the liner into position, and do not let go while it is still tilted or swinging.

Check: the plate lies flat and still, fully inside the plate outline, does not rock, does not touch the cutlery, and does not rest on the lip of the tray. If it lands off its outline, on the cutlery, or rocking, lift it clear, lay it back on the staging shelf, and place it again once.

Step 3: Place the two condiment pots

Goal: both condiment pots stand upright, each on its own labelled circle.

The left arm is the idle arm for the whole of Step 3. Read the circle label before each pick and take the pot that label names. Fill the **jam circle first, then the butter circle**, so nothing is carried over a pot already standing. One pot per trip, with the **right gripper** only.

- Read the label on the circle you are filling and find that pot on the staging shelf.
- Approach it from above and close on its body, square to it so it hangs upright, clear of its lid.
- Lift it straight up clear of the shelf and carry it upright to a point directly above its own circle, keeping it clear of the plate.
- Lower it straight down until it stands on the liner on its own base, hold it a moment until it is clearly standing on its own, then open the gripper and lift clear.

Grip the body and never the lid. Do not press down on the lid while lowering the pot, do not tip a pot on the carry, and do not stand one pot against the other to keep it upright.

Check: each pot stands upright and still on the circle its own name is on, fully inside that circle, lid unbroken, and clear of the other pot and of the plate. If a pot is on the wrong circle, lift it clear, stand it on the staging shelf, and place it again on the circle its own name is on. If it leans, rocks, or touches its neighbour, lift it clear and stand it down again once.

Step 4: Place the filled cup

Goal: the cup stands on its outline at the back right, still filled to its fill line, with no spill.

The left arm is the idle arm for the whole of Step 4. This is the last item onto the tray, and once it is down nothing passes over it again. Come in over the **back lip** of the tray, not over the condiment pots.

- The **right gripper** approaches the cup on the staging shelf from above and closes on its body, square to it so it hangs upright, clear of the rim and clear of the handle, with the handle pointing to the right.
- Lift it straight up clear of the shelf in a level carry and hold it at that height.
- Carry it in over the back lip of the tray to a point directly above the cup outline, keeping the rim flat and the juice surface still, and keeping the cup clear of the plate and both pots.

- Lower it straight down until it stands on the liner on its own base, hold it a moment until it is clearly standing on its own, then open the gripper and lift clear, straight up.

A full cup is top heavy. Move it slowly, keep it level from the lift to the release, and let it settle before the gripper opens. Never grip the rim or the handle, never turn the cup in the air to change where the handle points, and never top it up or pour any out.

If it comes up with the handle pointing the wrong way, stand it back on the staging shelf, let go, and take it again with the handle pointing to the right.

Check: the cup stands upright and still, fully inside the cup outline, handle pointing to the right, juice level with its fill line, and no juice anywhere outside the cup. If it rocks or leans, lift it clear in a level carry and stand it down again once. **Do not start Step 5 with a spill on the tray or with the cup off its outline.**

Step 5: Slide the tray to the delivery zone

Goal: the loaded tray sits square on the delivery outline with nothing moved on it and no spill.

Both grippers work together for the whole of Step 5, and there is no idle arm. The tray stays flat on the table the whole way. Neither arm passes over the loaded tray at any point: each comes in from its own end.

- The **left gripper** comes in from the left and closes on the **left rail**. The **right gripper** comes in from the right and closes on the **right rail**. Close both at the same time, square to the rails, with the tray still flat on the table.
- Slide the tray straight toward the front table edge in one continuous run, both grippers travelling together at the same speed, keeping the tray flat on the table and square to the front table edge.
- Slow the run as the tray comes onto the delivery outline and stop both grippers together once the tray sits fully inside the delivery outline.
- Open both grippers, lift both clear straight up, and return both arms to home.

Move the tray in one steady run. Do not lift it off the table, do not jerk it into motion, do not stop and restart part way, do not turn it, and do not push one end ahead of the other. Do not push the tray past the delivery outline or into the front table edge.

Check: the tray sits square on the delivery outline, flat on the table. Every item is still on its own mark, the cup still stands upright with its juice level with the fill line, the cutlery is still on the napkin, and there is no spill. If the tray stops short of the outline or ends crooked, take both rails again and slide it once more in one run. If an item has moved off its mark, lift that item clear with the gripper that owns it and set it down again once.

Step 6: End the episode

- Confirm the left end: the napkin lies flat on its outline and the fork and the spoon lie parallel on it, handles toward the front table edge.
- Confirm the center and right: the plate lies flat on its outline, both pots stand on their own labelled circles with lids unbroken, and the cup stands on its outline with its handle to the right.
- Confirm the cup: the juice is level with the fill line and no juice is anywhere outside the cup.
- Confirm the tray: it sits square and fully inside the delivery outline, flat on the table, with nothing overhanging it.
- Confirm the zones: the linen mat and the staging shelf are empty and the build spot is bare.
- Correct any failed check before ending.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the cup off the tray, empty it, and refill it to its fill line. Stand it on the staging shelf with its handle pointing to the right.
2. Take both condiment pots off the tray and stand them upright on the staging shelf. Take the plate off and lay it flat on the staging shelf.
3. Take the fork and the spoon off the napkin and lay them on their own spots on the linen mat. Take the napkin off the tray and lay it flat and folded on the linen mat.
4. Wipe and dry the liner and the tray, and wipe up any juice on the table, the shelf, or the linen mat.
5. Slide the empty tray back to the build spot, squared to the back table edge, with its liner lying flat inside it.
6. Replace a napkin that no longer holds its fold, a condiment pot with a broken or loosened lid, a cup that rocks when full, and a plate that rocks on a flat surface.
7. Replace a liner that is stained, curled, or torn so a mark cannot be read on camera.
8. Clear any debris off the table so the tray slides clean, and press down or replace any delivery outline tape that has lifted, curled, or torn, then run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an

annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the steps run out of order, a piece of cutlery goes down before the napkin, the spoon goes down before the fork, the butter pot goes down before the jam pot, the cup goes onto the tray before the plate or the pots, an item is carried over the cup once the cup is down, an item is carried over the napkin or the cutlery already set, the tray is slid before every item is on its own mark, or a step is skipped.
- **SOP rule broken:** Steps 1 to 5, napkin, cutlery, plate, condiments, cup, then the slide, filling the far mark of a pair first and carrying nothing over an item already down.
- **Coaching note:** napkin, cutlery, plate, condiments, cup, slide. The cup is always last, and once it is down nothing crosses over it.

Violation: Arms swapped roles

- **Visible cue:** the **left gripper** touches the plate, a condiment pot, or the cup; the **right gripper** touches the napkin, the fork, or the spoon; the left gripper reaches past the tray to the staging shelf or the right gripper reaches past the tray to the linen mat; the wrong gripper takes a rail in Step 5; or outside Step 5 the arm not named in a step comes over the tray, steadies an item, holds the napkin down, holds the tray, or assists the working arm.
- **SOP rule broken:** Steps 1 to 5, the left gripper handles the linen and the cutlery, the right gripper handles the plate, the pots, and the cup, and the arm not named in a step waits at home.
- **Coaching note:** left gripper takes linen, right gripper takes service, one arm over the tray at a time. Step 5 is the one shared action, left gripper on the left rail and right gripper on the right.

Violation: Napkin laid wrong

- **Visible cue:** the **left gripper** balls the napkin up, holds it by a corner so it hangs slack, unfolds it, refolds it, or shakes it out; or the napkin lands folded over on itself, crooked to the outline, partly off the napkin outline, over the lip of the tray, or on top of another mark.
- **SOP rule broken:** Step 1.1, carry the napkin flat with its fold kept and lay it flat and fully inside the napkin outline.
- **Coaching note:** take one edge square so it hangs flat, and lower it onto the outline rather than dropping it. The fold it arrives with is the fold it keeps.

Violation: Cutlery set wrong

- **Visible cue:** the **left gripper** lays the fork or the spoon on the bare liner, on the plate, on the lip of the tray, or half off the napkin; the two pieces touch, cross, or lie at visibly different angles; a handle points away from the front table edge; the fork and the spoon end up on each other's half of the napkin; or the left gripper turns a piece in the air to change the way it points.
- **SOP rule broken:** Steps 1.2 and 1.3, both pieces lie flat and parallel on the napkin, handles toward the front table edge, fork on the half nearer the plate and spoon on the half nearer the left rail.
- **Coaching note:** lift the piece already pointing the right way. If it comes up wrong, lay it back on the linen mat and take it again rather than turning it in the air.

Violation: Plate placed wrong

- **Visible cue:** the **right gripper** tips the plate on edge during the carry, slides it across the liner into position, or opens while the plate is still tilted or swinging; or the plate lands off the plate outline, on the cutlery or the napkin, on the lip of the tray, rocking, or overhanging the tray.
- **SOP rule broken:** Step 2, carry the plate level and lay it flat and fully inside the plate outline, releasing only once it rests on its own.
- **Coaching note:** grip the rim square so it hangs level, and set it down flat in one lowering. The plate is the widest item, so it lands before anything can be knocked by it.

Violation: Condiment pot placed wrong or lid disturbed

- **Visible cue:** the **right gripper** stands a pot on the circle the other pot is labelled for, or the two pots end up swapped; a pot leans, rocks, touches its neighbour or the plate, sits partly off its circle, or only stays upright because the other pot props it; or the gripper closes on a lid, presses a lid down, peels or lifts a lid, or a lid is left broken or loose.
- **SOP rule broken:** Step 3, read the circle label and take the pot it names, grip the body clear of the lid, and stand each pot upright and fully inside its own circle with its lid unbroken.
- **Coaching note:** read the label, then pick. Grip the body, because a lid taken as a handle is a lid that opens later.

Violation: Cup handled wrong

- **Visible cue:** the **right gripper** closes on the cup's rim or handle, tips or swings the cup on the carry, carries it high over the plate or the pots instead of in over the back lip, turns it in the air to move the handle, sets it down with the handle pointing anywhere but to the right, opens while the cup is still tilted or unsettled, or nudges it after release; or the cup lands off the cup outline, rocking, or leaning on the lip of the tray.

- **SOP rule broken:** Step 4, grip the body clear of the rim and handle, keep a level carry in over the back lip, and stand the cup upright and fully inside the cup outline with its handle to the right.
- **Coaching note:** body grip, one height, slow. A full cup is top heavy, so let it settle on the liner before the gripper opens.

Violation: Spill

- **Visible cue:** juice leaves the cup at any point, whether it slops on the lift, on the carry, on the set down, or during the slide; juice shows on the liner, the tray, an item, the shelf, or the table; a wet ring is left where the cup stood; the juice level ends visibly below the fill line; the cup is topped up or poured out; or condiment leaves a pot.
- **SOP rule broken:** Steps 4 and 5, keep the juice in the cup with a level carry and a steady slide, and never top up or pour out of the cup.
- **Coaching note:** the whole task is judged on the juice staying in the cup. Slow the lift, slow the set down, and slow the slide as the tray comes onto the delivery outline.

Violation: Tray not slid by both grippers together

- **Visible cue:** one gripper takes a rail while the other waits at home or hovers without closing, the second gripper joins only part way through the run, one gripper leads and the other trails, the two ends travel at visibly different speeds, or the tray goes over skewed so it arrives crooked.
- **SOP rule broken:** Step 5, the left gripper and the right gripper slide the tray together.
- **Coaching note:** both rails, closed at the same time, then start, travel, and stop as one. The tray stays square because both ends move the same amount.

Violation: Tray moved wrong

- **Visible cue:** the tray lifts off the table at any point in the run, is jerked into motion, is stopped and restarted part way, is turned, is pushed past the delivery outline or into the front table edge, or ends short of the outline, crooked to the front table edge, or partly outside it; or the tray is pushed, dragged, turned, or lifted during Steps 1 to 4.
- **SOP rule broken:** Steps 1 to 5, the tray stays flat on the table and still until Step 5, then goes to the delivery outline in one continuous run.
- **Coaching note:** flat on the table, one run, ease off at the end. The tray only moves once in the whole episode.

Violation: Bad pick, carry, or release

- **Visible cue:** two or more items are carried in one trip; an item is pushed, slid, or scraped across the shelf, the linen mat, or the liner instead of being lifted clear; an item is turned in the air to change the way it faces; a gripper opens over the bare liner, over another item, or over a mark the held item does not belong on; a gripper nudges, pats, or slides an item into

position after releasing it; or a failed check is answered with a recovery the SOP does not write.

- **SOP rule broken:** Steps 1 to 5, one item per trip, lifted clear of the surface, carried without turning it, released only over its own mark, a bad placement corrected by lifting the item clear and setting it down again, and each check answered only with the recovery it states.
- **Coaching note:** lift before you carry, and know the mark before the gripper opens. To change the way an item faces, set it back on its staging spot and take it again.

Violation: Dropped, collided, or knocked over

- **Visible cue:** an item falls or is struck; a pot topples on the shelf or on the liner; the cup tips; the plate is dropped onto the liner from above; an arm knocks an item already placed on its approach or its retract; an arm catches the lip of the tray on the way in; or an item is pushed off the tray or off the table.
- **SOP rule broken:** Steps 1 to 5, keep each item on a clear travel path, come onto the tray straight down from above, and release only after a settled placement.
- **Coaching note:** check the path and the mark before the arm moves, and retract straight up so the gripper does not catch a standing pot or the cup.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an item still on the linen mat or the staging shelf, an empty mark on the liner, two items touching, cutlery off the napkin, the cup off its outline or below its fill line, a spill anywhere, the tray still on the build spot or crooked on the delivery outline, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 6, confirm the left end, the center and right, the cup, the tray, and the zones, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective cup:** one that rocks when it is full to its fill line, leaks, or slips out of a correct body grip.
- **Defective condiment pot:** one whose lid is already broken or loose at episode start, or that will not stand upright unaided on a bare flat surface.

- **Defective plate:** one that is chipped, or that rocks on a flat surface, or that slips out of a correct rim grip.
- **Defective napkin:** one that will not hold its fold when it is lifted by one edge, or that is damp at episode start.
- **Defective cutlery:** a fork or spoon that rocks on a flat surface or rolls off the napkin after a correct flat placement.
- **Tray will not slide:** it catches, tips, or skews under a correct two rail push because the table is dusty, slick, or uneven.
- **Liner will not stay put:** it slides or lifts inside the tray when an item is set down straight from above, or a mark cannot be read on camera.
- **A zone shifts during the episode,** or a zone turns out to sit outside its arm's comfortable reach so a mark, a staging spot, or a rail cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Lay the folded napkin on the tray
2. Place the fork on the napkin
3. Place the spoon on the napkin
4. Place the plate on the tray
5. Place one condiment pot on its circle
6. Place the filled cup on the tray
7. Slide the loaded tray to the delivery zone
8. Return both arms home and end the episode